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CSMAP-517 C#.NET and Artificial Intelligence Laboratory

C#.NET PROGRAM SOLUTIONS

Part A — Console Applications

1. Write a C# Console Application program to calculate the factorial of a given number.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        long factorial = 1;
        for (int i = 1; i <= n; i++)
            factorial *= i;

        Console.WriteLine("Factorial = " + factorial);
    }
}
```

2. Write a C# Console Application program to display the multiplication table of a given number.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        for (int i = 1; i <= 10; i++)
            Console.WriteLine(n + " x " + i + " = " + (n * i));
    }
}
```

3. Write a C# Console Application program to reverse the digits of a given number using recursive function.

```
using System;

class Program
{
    static int ReverseNumber(int n, int rev = 0)
    {
```

```
        if (n == 0)
            return rev;

        return ReverseNumber(n / 10, rev * 10 + n % 10);
    }

    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int number = Math.Abs(n);
        int reverse = ReverseNumber(number);

        if (n < 0)
            reverse = -reverse;

        Console.WriteLine("Reverse = " + reverse);
    }
}
```

4. Write a C# Console Application program to check whether a given number is Prime.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());
        bool prime = n >= 2;

        for (int i = 2; i * i <= n && prime; i++)
        {
            if (n % i == 0)
                prime = false;
        }

        Console.WriteLine(prime ? "Prime Number" : "Not a Prime Number");
    }
}
```

5. Write a C# Console Application program to check whether a given number is an Armstrong number or not.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int temp = Math.Abs(n);
        int digits = temp.ToString().Length;
        int sum = 0;

        while (temp > 0)
        {
```

```
            int digit = temp % 10;
            sum += (int)Math.Pow(digit, digits);
            temp /= 10;
        }

        Console.WriteLine(sum == Math.Abs(n)
            ? "Armstrong Number"
            : "Not an Armstrong Number");
    }
}
```

6. Write a C# Console Application program to check whether a given number is a palindrome.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int original = n;
        int temp = Math.Abs(n);
        int reverse = 0;

        while (temp > 0)
        {
            reverse = reverse * 10 + temp % 10;
            temp /= 10;
        }

        Console.WriteLine(Math.Abs(original) == reverse
            ? "Palindrome Number"
            : "Not a Palindrome Number");
    }
}
```

7. Write a C# Console Application program to calculate the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of two numbers.

```
using System;

class Program
{
    static int GCD(int a, int b)
    {
        a = Math.Abs(a);
        b = Math.Abs(b);

        while (b != 0)
        {
            int r = a % b;
            a = b;
            b = r;
        }
        return a;
    }

    static void Main()
```

```
    {
        Console.Write("Enter first number: ");
        int a = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter second number: ");
        int b = Convert.ToInt32(Console.ReadLine());

        int gcd = GCD(a, b);
        long lcm = (a == 0 || b == 0) ? 0 : Math.Abs((long)a * b) / gcd;

        Console.WriteLine("GCD = " + gcd);
        Console.WriteLine("LCM = " + lcm);
    }
}
```

8. Write a C# Console Application program to accept a number from the user and convert it into binary, octal and hexadecimal using switch case.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int n = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("1. Binary");
        Console.WriteLine("2. Octal");
        Console.WriteLine("3. Hexadecimal");
        Console.Write("Enter choice: ");
        int choice = Convert.ToInt32(Console.ReadLine());

        switch (choice)
        {
            case 1:
                Console.WriteLine("Binary = " + Convert.ToString(n, 2));
                break;
            case 2:
                Console.WriteLine("Octal = " + Convert.ToString(n, 8));
                break;
            case 3:
                Console.WriteLine("Hexadecimal = " + Convert.ToString(n, 16).ToUpper());
                break;
            default:
                Console.WriteLine("Invalid choice");
                break;
        }
    }
}
```

9. Write a C# Console Application program to sort an array in ascending and descending order.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter number of elements: ");
```

```
        int n = Convert.ToInt32(Console.ReadLine());

        int[] a = new int[n];

        for (int i = 0; i < n; i++)
        {
            Console.Write("Enter element " + (i + 1) + ": ");
            a[i] = Convert.ToInt32(Console.ReadLine());
        }

        Array.Sort(a);

        Console.WriteLine("Ascending Order:");
        Console.WriteLine(string.Join(" ", a));

        Array.Reverse(a);

        Console.WriteLine("Descending Order:");
        Console.WriteLine(string.Join(" ", a));
    }
}
```

10. Write a C# Console Application program to find the largest and smallest number from an array.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter number of elements: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int[] a = new int[n];

        for (int i = 0; i < n; i++)
        {
            Console.Write("Enter element " + (i + 1) + ": ");
            a[i] = Convert.ToInt32(Console.ReadLine());
        }

        int largest = a[0];
        int smallest = a[0];

        for (int i = 1; i < n; i++)
        {
            if (a[i] > largest) largest = a[i];
            if (a[i] < smallest) smallest = a[i];
        }

        Console.WriteLine("Largest = " + largest);
        Console.WriteLine("Smallest = " + smallest);
    }
}
```

11. Write a C# Console Application program to accept one string & a character, find the occurrences of character in String using function.

```
using System;
```

```
class Program
{
    static int CountCharacter(string text, char ch)
    {
        int count = 0;

        foreach (char c in text)
        {
            if (char.ToLower(c) == char.ToLower(ch))
                count++;
        }

        return count;
    }

    static void Main()
    {
        Console.Write("Enter a string: ");
        string text = Console.ReadLine();

        Console.Write("Enter a character: ");
        char ch = Convert.ToChar(Console.ReadLine());

        Console.WriteLine("Occurrences = " + CountCharacter(text, ch));
    }
}
```

12. Write a C# Console Application program to calculate the sum of first and last digits of a given number.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        long n = Math.Abs(Convert.ToInt64(Console.ReadLine()));

        long last = n % 10;
        long first = n;

        while (first >= 10)
            first /= 10;

        Console.WriteLine("Sum of first and last digits = " + (first + last));
    }
}
```

13. Write a C# Console Application program that prompts the user to input a numeric integer and throws an exception if the number is less than 0 or greater than 1000.

```
using System;

class NumberOutOfRangeException : Exception

{

public NumberOutOfRangeException(string message)

: base(message)
```

```
{
}
}
class Program
{
static void Main()
{
try
{
Console.Write("Enter a number: ");
int n = Convert.ToInt32(Console.ReadLine());
if (n < 0 || n > 1000)
{
throw new NumberOutOfRangeException(
"Number must be between 0 and 1000."
);
}
Console.WriteLine("Valid Number: " + n);
}
catch (NumberOutOfRangeException ex)
{
Console.WriteLine("User Defined Exception: " + ex.Message);
}
catch (FormatException)
{
Console.WriteLine("Please enter a valid integer.");
}
Console.ReadKey();
}
}
```

14. Write a C# Console Application program to accept a user name from user and greets based on current time. (Use Switch case)

```
using System;

class Program
{
    static void Main()
```

```
    {
        Console.Write("Enter your name: ");
        string name = Console.ReadLine();

        int hour = DateTime.Now.Hour;
        string greeting;

        switch (hour / 6)
        {
            case 0:
                greeting = "Good Morning";
                break;
            case 1:
                greeting = "Good Afternoon";
                break;
            case 2:
                greeting = "Good Evening";
                break;
            default:
                greeting = "Good Night";
                break;
        }

        Console.WriteLine(greeting + ", " + name + "!");
    }
}
```

15. Write a C# Console Application program to perform arithmetic operations on two numbers. Define methods for addition, subtraction, multiplication, and division. Create a delegate that can reference these methods and allow the user to select an operation at runtime. Display the result of the selected operation.

```
using System;

class Program
{
    delegate double Operation(double a, double b);

    static double Add(double a, double b) => a + b;
    static double Subtract(double a, double b) => a - b;
    static double Multiply(double a, double b) => a * b;

    static double Divide(double a, double b)
    {
        if (b == 0)
            throw new DivideByZeroException("Cannot divide by zero.");
        return a / b;
    }

    static void Main()
    {
        Console.Write("Enter first number: ");
        double a = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter second number: ");
        double b = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("1. Addition");
        Console.WriteLine("2. Subtraction");
        Console.WriteLine("3. Multiplication");
        Console.WriteLine("4. Division");
```

```
        Console.Write("Enter choice: ");
        int choice = Convert.ToInt32(Console.ReadLine());

        Operation op;

        switch (choice)
        {
            case 1: op = Add; break;
            case 2: op = Subtract; break;
            case 3: op = Multiply; break;
            case 4: op = Divide; break;
            default:
                Console.WriteLine("Invalid choice.");
                return;
        }

        try
        {
            Console.WriteLine("Result = " + op(a, b));
        }
        catch (DivideByZeroException ex)
        {
            Console.WriteLine(ex.Message);
        }
    }
}
```

16. Write a C# Console Application program to perform different string operations. The application should accept a string from the user and provide options to find the length of the string, convert it to uppercase, convert it to lowercase, reverse the string, and count the number of vowels. Create a delegate that can reference these methods. Define separate methods for each operation and use a delegate to invoke the selected operation.

```
using System;

class Program
{
    delegate string StringOperation(string text);

    static string Length(string text) => "Length = " + text.Length;
    static string Upper(string text) => text.ToUpper();
    static string Lower(string text) => text.ToLower();

    static string Reverse(string text)
    {
        char[] chars = text.ToCharArray();
        Array.Reverse(chars);
        return new string(chars);
    }

    static string CountVowels(string text)
    {
        int count = 0;
        foreach (char c in text.ToLower())
            if ("aeiou".Contains(c))
                count++;

        return "Number of vowels = " + count;
    }
```

```
    static void Main()
    {
        Console.Write("Enter a string: ");
        string text = Console.ReadLine();

        Console.WriteLine("1. Length");
        Console.WriteLine("2. Uppercase");
        Console.WriteLine("3. Lowercase");
        Console.WriteLine("4. Reverse");
        Console.WriteLine("5. Count vowels");
        Console.Write("Enter choice: ");
        int choice = Convert.ToInt32(Console.ReadLine());

        StringOperation operation;

        switch (choice)
        {
            case 1: operation = Length; break;
            case 2: operation = Upper; break;
            case 3: operation = Lower; break;
            case 4: operation = Reverse; break;
            case 5: operation = CountVowels; break;
            default:
                Console.WriteLine("Invalid choice.");
                return;
        }

        Console.WriteLine("Result = " + operation(text));
    }
}
```

17. Write a C# Console Application program to create a Student class containing appropriate data members and methods. Accept student details, calculate the total and percentage of marks, and display the complete student information using an object of the class.

```
using System;

class Student
{
    int rollNo;
    string name;
    double m1, m2, m3;

    public void Accept()
    {
        Console.Write("Enter Roll No: ");
        rollNo = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Name: ");
        name = Console.ReadLine();

        Console.Write("Enter marks for Subject 1: ");
        m1 = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter marks for Subject 2: ");
        m2 = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter marks for Subject 3: ");
        m3 = Convert.ToDouble(Console.ReadLine());
```

```
    }

    public double Total()
    {
        return m1 + m2 + m3;
    }

    public double Percentage()
    {
        return Total() / 3.0;
    }

    public void Display()
    {
        Console.WriteLine("\nStudent Information");
        Console.WriteLine("Roll No: " + rollNo);
        Console.WriteLine("Name: " + name);
        Console.WriteLine("Total: " + Total());
        Console.WriteLine("Percentage: " + Percentage().ToString("F2") + "%");
    }
}

class Program
{
    static void Main()
    {
        Student s = new Student();
        s.Accept();
        s.Display();
    }
}
```

18. Write a C# Console Application program to demonstrate interfaces by defining an IShape interface containing methods to calculate area and perimeter. Implement the interface in Circle and Rectangle classes and display the calculated results.

```
using System;

interface IShape
{
    double CalculateArea();
    double CalculatePerimeter();
}

class Circle : IShape
{
    double radius;

    public Circle(double radius)
    {
        this.radius = radius;
    }

    public double CalculateArea()
    {
        return Math.PI * radius * radius;
    }

    public double CalculatePerimeter()
    {
```

```
        return 2 * Math.PI * radius;
    }
}

class Rectangle : IShape
{
    double length, width;

    public Rectangle(double length, double width)
    {
        this.length = length;
        this.width = width;
    }

    public double CalculateArea()
    {
        return length * width;
    }

    public double CalculatePerimeter()
    {
        return 2 * (length + width);
    }
}

class Program
{
    static void Main()
    {
        Console.Write("Enter circle radius: ");
        double r = Convert.ToDouble(Console.ReadLine());

        Circle c = new Circle(r);

        Console.WriteLine("Circle Area = " + c.CalculateArea());
        Console.WriteLine("Circle Perimeter = " + c.CalculatePerimeter());

        Console.Write("\nEnter rectangle length: ");
        double l = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter rectangle width: ");
        double w = Convert.ToDouble(Console.ReadLine());

        Rectangle rect = new Rectangle(l, w);

        Console.WriteLine("Rectangle Area = " + rect.CalculateArea());
        Console.WriteLine("Rectangle Perimeter = " + rect.CalculatePerimeter());
    }
}
```

19. Write a C# Console Application program to demonstrate method hiding. Create a base class Employee with a Display() method and derive Manager and hide the base class method using the new keyword. Demonstrate the difference between the base class and derived class methods.

```
using System;

class Employee
{
    public void Display()
    {
```

```
        Console.WriteLine("Display() method of Employee class");
    }
}

class Manager : Employee
{
    public new void Display()
    {
        Console.WriteLine("Display() method of Manager class");
    }
}

class Program
{
    static void Main()
    {
        Employee employee = new Employee();
        employee.Display();

        Manager manager = new Manager();
        manager.Display();

        Employee reference = new Manager();
        reference.Display();

        Console.WriteLine("\nThe base-class reference calls Employee.Display() because the method is
hidden, not overridden.");
    }
}
```

20. Write a C# Console Application program to read the content of a file using StreamReader class.

```
using System;
using System.IO;

class Program
{
    static void Main()
    {
        Console.Write("Enter file path: ");
        string path = Console.ReadLine();

        try
        {
            using (StreamReader reader = new StreamReader(path))
            {
                string content = reader.ReadToEnd();
                Console.WriteLine("\nFile Content:");
                Console.WriteLine(content);
            }
        }
        catch (FileNotFoundException)
        {
            Console.WriteLine("File not found.");
        }
        catch (IOException ex)
        {
            Console.WriteLine("File error: " + ex.Message);
        }
    }
}
```

21. Write a C# Console Application program to demonstrate method overriding. Create a base class Shape with a virtual CalculateArea() method and override it in Circle and Rectangle classes to calculate their respective areas.

```
using System;

class Shape
{
    public virtual double CalculateArea()
    {
        return 0;
    }
}

class Circle : Shape
{
    double radius;

    public Circle(double radius)
    {
        this.radius = radius;
    }

    public override double CalculateArea()
    {
        return Math.PI * radius * radius;
    }
}

class Rectangle : Shape
{
    double length, width;

    public Rectangle(double length, double width)
    {
        this.length = length;
        this.width = width;
    }

    public override double CalculateArea()
    {
        return length * width;
    }
}

class Program
{
    static void Main()
    {
        Shape s;

        s = new Circle(5);
        Console.WriteLine("Circle Area = " + s.CalculateArea());

        s = new Rectangle(10, 5);
        Console.WriteLine("Rectangle Area = " + s.CalculateArea());
    }
}
```

22. Write a C# Console Application program to demonstrate runtime polymorphism using method overriding. Create a base class Vehicle with a virtual Start() method and override it in Car, Bike, and Bus classes. Invoke the appropriate method using a base class reference.

```
using System;

class Vehicle
{
    public virtual void Start()
    {
        Console.WriteLine("Vehicle is starting.");
    }
}

class Car : Vehicle
{
    public override void Start()
    {
        Console.WriteLine("Car starts with a key/button.");
    }
}

class Bike : Vehicle
{
    public override void Start()
    {
        Console.WriteLine("Bike starts with a key/button.");
    }
}

class Bus : Vehicle
{
    public override void Start()
    {
        Console.WriteLine("Bus starts with a key.");
    }
}

class Program
{
    static void Main()
    {
        Vehicle vehicle;

        vehicle = new Car();
        vehicle.Start();

        vehicle = new Bike();
        vehicle.Start();

        vehicle = new Bus();
        vehicle.Start();
    }
}
```

23. Write a C# Console Application program to read the content of a file in reverse order using StreamReader class.

```
using System;
```

```
using System.IO;

class Program
{
    static void Main()
    {
        Console.Write("Enter file path: ");
        string path = Console.ReadLine();

        try
        {
            using (StreamReader reader = new StreamReader(path))
            {
                string content = reader.ReadToEnd();

                char[] chars = content.ToCharArray();
                Array.Reverse(chars);

                Console.WriteLine("\nFile Content in Reverse Order:");
                Console.WriteLine(new string(chars));
            }
        }
        catch (FileNotFoundException)
        {
            Console.WriteLine("File not found.");
        }
        catch (IOException ex)
        {
            Console.WriteLine("File error: " + ex.Message);
        }
    }
}
```

24. Write a C# Console Application program to accept M x N matrix and perform Transpose of it.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter number of rows (M): ");
        int m = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter number of columns (N): ");
        int n = Convert.ToInt32(Console.ReadLine());

        int[,] matrix = new int[m, n];

        Console.WriteLine("Enter matrix elements:");

        for (int i = 0; i < m; i++)
            for (int j = 0; j < n; j++)
            {
                Console.Write("Element [" + i + "," + j + "]: ");
                matrix[i, j] = Convert.ToInt32(Console.ReadLine());
            }

        Console.WriteLine("\nTranspose:");

        for (int j = 0; j < n; j++)
```

```
        {
            for (int i = 0; i < m; i++)
                Console.Write(matrix[i, j] + " ");
            Console.WriteLine();
        }
    }
}
```

25. Write a C# Console Application program to accept two matrices and print addition of these two matrices.

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter rows: ");
        int rows = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter columns: ");
        int cols = Convert.ToInt32(Console.ReadLine());

        int[,] a = new int[rows, cols];
        int[,] b = new int[rows, cols];
        int[,] sum = new int[rows, cols];

        Console.WriteLine("Enter first matrix:");
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                a[i, j] = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter second matrix:");
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                b[i, j] = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("\nAddition:");
        for (int i = 0; i < rows; i++)
        {
            for (int j = 0; j < cols; j++)
            {
                sum[i, j] = a[i, j] + b[i, j];
                Console.Write(sum[i, j] + " ");
            }
            Console.WriteLine();
        }
    }
}
```

Part B — Windows Forms Applications

1. Create a Windows Forms application to perform addition, subtraction, multiplication, and division of two numbers. (Use Label, TextBox, Button Controls)

```
// Controls: txtNum1, txtNum2, btnAdd, btnSub, btnMul, btnDiv, lblResult

private void btnAdd_Click(object sender, EventArgs e)
{
    double a = Convert.ToDouble(txtNum1.Text);
    double b = Convert.ToDouble(txtNum2.Text);
    lblResult.Text = "Result = " + (a + b);
}

private void btnSub_Click(object sender, EventArgs e)
{
    double a = Convert.ToDouble(txtNum1.Text);
    double b = Convert.ToDouble(txtNum2.Text);
    lblResult.Text = "Result = " + (a - b);
}

private void btnMul_Click(object sender, EventArgs e)
{
    double a = Convert.ToDouble(txtNum1.Text);
    double b = Convert.ToDouble(txtNum2.Text);
    lblResult.Text = "Result = " + (a * b);
}

private void btnDiv_Click(object sender, EventArgs e)
{
    double a = Convert.ToDouble(txtNum1.Text);
    double b = Convert.ToDouble(txtNum2.Text);

    if (b == 0)
        lblResult.Text = "Cannot divide by zero.";
    else
        lblResult.Text = "Result = " + (a / b);
}
```

2. Write a C# program to create a windows form application to display a list of colors in ComboBox. Add a button on the form, when the user clicks the button then the color of the form must be changed to the selected color from the ComboBox.

```
// Controls: cmbColor, btnChangeColor

private void Form1_Load(object sender, EventArgs e)
{
    cmbColor.Items.Add("Red");
    cmbColor.Items.Add("Green");
    cmbColor.Items.Add("Blue");
    cmbColor.Items.Add("Yellow");
    cmbColor.Items.Add("White");
}

private void btnChangeColor_Click(object sender, EventArgs e)
```

```
{
    switch (cmbColor.Text)
    {
        case "Red": this.BackColor = Color.Red; break;
        case "Green": this.BackColor = Color.Green; break;
        case "Blue": this.BackColor = Color.Blue; break;
        case "Yellow": this.BackColor = Color.Yellow; break;
        case "White": this.BackColor = Color.White; break;
    }
}
```

3. Create a Windows Forms application that accepts the student name and displays "Welcome, [Name]" when the button is clicked. (Use Label, TextBox, Button Controls)

```
// Controls: txtName, btnWelcome, lblMessage

private void btnWelcome_Click(object sender, EventArgs e)
{
    string name = txtName.Text.Trim();

    if (name == "")
        lblMessage.Text = "Please enter student name.";
    else
        lblMessage.Text = "Welcome, " + name;
}
```

4. Create a Windows Forms application to select hobbies (Reading, Music, Sports, Traveling) and display the selected hobbies. (Use CheckBox, Button, Label Controls)

```
// Controls: chkReading, chkMusic, chkSports, chkTraveling, btnDisplay, lblHobbies

private void btnDisplay_Click(object sender, EventArgs e)
{
    string hobbies = "";

    if (chkReading.Checked) hobbies += "Reading, ";
    if (chkMusic.Checked) hobbies += "Music, ";
    if (chkSports.Checked) hobbies += "Sports, ";
    if (chkTraveling.Checked) hobbies += "Traveling, ";

    if (hobbies.Length == 0)
        lblHobbies.Text = "No hobby selected.";
    else
        lblHobbies.Text = "Selected Hobbies: " +
                          hobbies.TrimEnd(',', ' ');
}
```

5. Create a Windows Forms application to select a country from a ComboBox and display its capital. (Use ComboBox, Button, Label Controls)

```
// Controls: cmbCountry, btnCapital, lblCapital

private void Form1_Load(object sender, EventArgs e)
{
    cmbCountry.Items.Add("India");
    cmbCountry.Items.Add("Japan");
    cmbCountry.Items.Add("France");
    cmbCountry.Items.Add("United Kingdom");
    cmbCountry.Items.Add("Australia");
```

```
}

private void btnCapital_Click(object sender, EventArgs e)
{
    switch (cmbCountry.Text)
    {
        case "India": lblCapital.Text = "Capital: New Delhi"; break;
        case "Japan": lblCapital.Text = "Capital: Tokyo"; break;
        case "France": lblCapital.Text = "Capital: Paris"; break;
        case "United Kingdom": lblCapital.Text = "Capital: London"; break;
        case "Australia": lblCapital.Text = "Capital: Canberra"; break;
        default: lblCapital.Text = "Please select a country."; break;
    }
}
```

6. Create a Windows Forms application that accepts date of birth from the user and calculates the age. (Use DateTimePicker, Label, Button Controls).

```
// Controls: dtpDOB, btnCalculate, lblAge

private void btnCalculate_Click(object sender, EventArgs e)
{
    DateTime dob = dtpDOB.Value.Date;
    DateTime today = DateTime.Today;

    int age = today.Year - dob.Year;

    if (dob > today.AddYears(-age))
        age--;

    if (dob > today)
        lblAge.Text = "Invalid date of birth.";
    else
        lblAge.Text = "Age = " + age + " years";
}
```

7. Create a Windows Forms application that changes the font size of a label "COMPUTER SCIENCE" using a TrackBar. (Use TrackBar, Label Controls)

```
// Controls: lblText, trackBarFontSize
// Set lblText.Text = "COMPUTER SCIENCE"
// Set TrackBar Minimum = 8, Maximum = 48

private void trackBarFontSize_Scroll(object sender, EventArgs e)
{
    lblText.Font = new Font(
        lblText.Font.FontFamily,
        trackBarFontSize.Value,
        lblText.Font.Style);
}
```

8. Create a Windows Forms application that accepts a number from the user and converts number to word. (Use MaskedTextBox to validate a number. Example: If no is 345 then output should be Three Four Five)

```
// Controls: maskedTextBox1, btnConvert, lblResult

private string DigitToWord(char digit)
{
    switch (digit)
```

```
    {
        case '0': return "Zero";
        case '1': return "One";
        case '2': return "Two";
        case '3': return "Three";
        case '4': return "Four";
        case '5': return "Five";
        case '6': return "Six";
        case '7': return "Seven";
        case '8': return "Eight";
        case '9': return "Nine";
        default: return "";
    }
}

private void btnConvert_Click(object sender, EventArgs e)
{
    string number = maskedTextBox1.Text.Trim();

    if (!long.TryParse(number, out _))
    {
        lblResult.Text = "Enter a valid number.";
        return;
    }

    string result = "";

    foreach (char c in number)
    {
        if (char.IsDigit(c))
            result += DigitToWord(c) + " ";
    }

    lblResult.Text = result.Trim();
}
```

9. Create a windows application to accept login name and password from the user. If Login name equals to password display successful message to user else accept login name and password.

```
// Controls: txtLoginName, txtPassword, btnLogin, lblMessage
// Set txtPassword.UseSystemPasswordChar = true

private void btnLogin_Click(object sender, EventArgs e)
{
    string loginName = txtLoginName.Text;
    string password = txtPassword.Text;

    if (loginName == password)
        lblMessage.Text = "Login Successful";
    else
        lblMessage.Text = "Login Name and Password are not equal.";
}
```

10. Create a windows application to change the text entered in the Textbox according to the selected font, font color and font size.

```
// Controls: txtText, cmbFont, cmbSize, btnColor, btnApply
// A ColorDialog named colorDialog1 is also used.

private void Form1_Load(object sender, EventArgs e)
```

```
{
    foreach (FontFamily font in FontFamily.Families)
        cmbFont.Items.Add(font.Name);

    for (int i = 8; i <= 48; i += 2)
        cmbSize.Items.Add(i);

    cmbFont.SelectedItem = "Arial";
    cmbSize.SelectedItem = 12;
}

private void btnColor_Click(object sender, EventArgs e)
{
    if (colorDialog1.ShowDialog() == DialogResult.OK)
        txtText.ForeColor = colorDialog1.Color;
}

private void btnApply_Click(object sender, EventArgs e)
{
    string fontName = cmbFont.Text;
    float size = Convert.ToSingle(cmbSize.Text);

    txtText.Font = new Font(fontName, size);
}
```

11. Create a windows application which contains a menu “File” which contains sub-menus like “New (Ctrl+N)”, “Open (Ctrl+O)”, “Save (Ctrl+S)” and RichTextBox.

```
// Controls: MenuStrip with File menu and New/Open/Save items;
// RichTextBox named richTextBox1.

private void newToolStripMenuItem_Click(object sender, EventArgs e)
{
    richTextBox1.Clear();
}

private void openToolStripMenuItem_Click(object sender, EventArgs e)
{
    using (OpenFileDialog open = new OpenFileDialog())
    {
        open.Filter = "Text Files|*.txt|All Files|*.*";

        if (open.ShowDialog() == DialogResult.OK)
            richTextBox1.Text = File.ReadAllText(open.FileName);
    }
}

private void saveToolStripMenuItem_Click(object sender, EventArgs e)
{
    using (SaveFileDialog save = new SaveFileDialog())
    {
        save.Filter = "Text Files|*.txt|All Files|*.*";

        if (save.ShowDialog() == DialogResult.OK)
            File.WriteAllText(save.FileName, richTextBox1.Text);
    }
}

// Set ShortcutKeys in the Designer:
// New  -> Ctrl+N
```

```
// Open -> Ctrl+O
// Save -> Ctrl+S
```

12. Create a windows application which accepts string from user & counts total no of vowels, occurrence of each vowel.

```
// Controls: txtInput, btnCount, lblTotal, lblA, lblE, lblI, lblO, lblU

private void btnCount_Click(object sender, EventArgs e)
{
    string text = txtInput.Text.ToLower();

    int a = 0, e1 = 0, i = 0, o = 0, u = 0;

    foreach (char ch in text)
    {
        switch (ch)
        {
            case 'a': a++; break;
            case 'e': e1++; break;
            case 'i': i++; break;
            case 'o': o++; break;
            case 'u': u++; break;
        }
    }

    int total = a + e1 + i + o + u;

    lblTotal.Text = "Total Vowels = " + total;
    lblA.Text = "A = " + a;
    lblE.Text = "E = " + e1;
    lblI.Text = "I = " + i;
    lblO.Text = "O = " + o;
    lblU.Text = "U = " + u;
}
```

13. Create a windows application which converts given decimal no to binary, octal and hexadecimal.

```
// Controls: txtDecimal, btnConvert, lblBinary, lblOctal, lblHexadecimal

private void btnConvert_Click(object sender, EventArgs e)
{
    if (!int.TryParse(txtDecimal.Text, out int n) || n < 0)
    {
        MessageBox.Show("Enter a valid non-negative integer.");
        return;
    }

    lblBinary.Text = "Binary = " + Convert.ToString(n, 2);
    lblOctal.Text = "Octal = " + Convert.ToString(n, 8);
    lblHexadecimal.Text = "Hexadecimal = " +
                          Convert.ToString(n, 16).ToUpper();
}
```

14. Create a windows application which accepts 2 numbers from the user and computes GCD and LCM of 2 numbers.

```
// Controls: txtNum1, txtNum2, btnCalculate, lblGCD, lblLCM

private int GCD(int a, int b)
```

```
{
    a = Math.Abs(a);
    b = Math.Abs(b);

    while (b != 0)
    {
        int r = a % b;
        a = b;
        b = r;
    }

    return a;
}

private void btnCalculate_Click(object sender, EventArgs e)
{
    int a = Convert.ToInt32(txtNum1.Text);
    int b = Convert.ToInt32(txtNum2.Text);

    int gcd = GCD(a, b);
    long lcm = (a == 0 || b == 0) ? 0 :
               Math.Abs((long)a * b) / gcd;

    lblGCD.Text = "GCD = " + gcd;
    lblLCM.Text = "LCM = " + lcm;
}
```

15. Create a C# Windows Forms application using ADO.NET to insert student details into a MS Access database. Consider a table named Student with the attributes as RollNo, Name, Course,and Marks. The application should provide suitable controls to accept the four values and an Insert button to store the record in the database. Display an appropriate success/error message after insertion.

```
// Controls: txtRollNo, txtName, txtCourse, txtMarks, btnInsert
// Database: College.accdb
// Table: Student(RollNo, Name, Course, Marks)
//
// Add reference/namespace: System.Data.OleDb

using System.Data.OleDb;

string connectionString =
    @"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=College.accdb;";

private void btnInsert_Click(object sender, EventArgs e)
{
    try
    {
        using (OleDbConnection con =
               new OleDbConnection(connectionString))
        {
            string query =
                "INSERT INTO Student (RollNo, Name, Course, Marks) " +
                "VALUES (?, ?, ?, ?)";

            using (OleDbCommand cmd = new OleDbCommand(query, con))
            {
                cmd.Parameters.AddWithValue("@p1",
                    Convert.ToInt32(txtRollNo.Text));
                cmd.Parameters.AddWithValue("@p2", txtName.Text);
                cmd.Parameters.AddWithValue("@p3", txtCourse.Text);
```

```
                cmd.Parameters.AddWithValue("@p4",
                    Convert.ToDouble(txtMarks.Text));

                con.Open();
                cmd.ExecuteNonQuery();

                MessageBox.Show("Student record inserted successfully.");
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
```

16. Create a C# Windows Forms application using ADO.NET and DataGridView to retrieve and display all student records from the Student table of the MS Access database.

```
// Controls: DataGridView named dataGridView1 and Button named btnLoad
// Database: College.accdb
// Add namespace: System.Data.OleDb

using System.Data;
using System.Data.OleDb;

string connectionString =
    @"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=College.accdb;";

private void btnLoad_Click(object sender, EventArgs e)
{
    try
    {
        using (OleDbConnection con =
               new OleDbConnection(connectionString))
        {
            string query = "SELECT * FROM Student";

            using (OleDbDataAdapter adapter =
                   new OleDbDataAdapter(query, con))
            {
                DataTable table = new DataTable();
                adapter.Fill(table);

                dataGridView1.DataSource = table;
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
```

17. Create a C# Windows Forms application using ADO.NET to insert employee details into a MS Access database. Consider a table named Employee with the attributes as EmpId, Name, Department, and Salary. The application should provide suitable controls to accept the four values and an Insert button to store the record in the table. Display an appropriate success/error message after insertion.

```
// Controls: txtEmpId, txtName, txtDepartment, txtSalary, btnInsert
// Database: College.accdb
// Table: Employee(EmpId, Name, Department, Salary)
//
// Add namespace: System.Data.OleDb

using System.Data.OleDb;

string connectionString =
    @"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=College.accdb;";

private void btnInsert_Click(object sender, EventArgs e)
{
    try
    {
        using (OleDbConnection con =
               new OleDbConnection(connectionString))
        {
            string query =
                "INSERT INTO Employee " +
                "(EmpId, Name, Department, Salary) " +
                "VALUES (?, ?, ?, ?)";

            using (OleDbCommand cmd = new OleDbCommand(query, con))
            {
                cmd.Parameters.AddWithValue("@p1",
                    Convert.ToInt32(txtEmpId.Text));
                cmd.Parameters.AddWithValue("@p2", txtName.Text);
                cmd.Parameters.AddWithValue("@p3", txtDepartment.Text);
                cmd.Parameters.AddWithValue("@p4",
                    Convert.ToDouble(txtSalary.Text));

                con.Open();
                cmd.ExecuteNonQuery();

                MessageBox.Show("Employee record inserted successfully.");
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
```

18. Create a C# Windows Forms application using ADO.NET and DataGridView to retrieve and display all employee records from the Employee table from MS Access Database.

```
// Controls: DataGridView named dataGridView1 and Button named btnLoad
// Database: College.accdb
// Add namespace: System.Data.OleDb

using System.Data;
using System.Data.OleDb;

string connectionString =
    @"Provider=Microsoft.ACE.OLEDB.12.0;Data Source=College.accdb;";

private void btnLoad_Click(object sender, EventArgs e)
{
```

```
    try
    {
        using (OleDbConnection con =
               new OleDbConnection(connectionString))
        {
            string query = "SELECT * FROM Employee";

            using (OleDbDataAdapter adapter =
                   new OleDbDataAdapter(query, con))
            {
                DataTable table = new DataTable();
                adapter.Fill(table);

                dataGridView1.DataSource = table;
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
```

19. Create a Windows Forms application to select Languages (Marathi,English, Hindi, Sanskrit, Tamil) and display the selected languages. (Use CheckBox, Button, Label Controls)

```
// Controls:
// chkMarathi, chkEnglish, chkHindi, chkSanskrit, chkTamil
// btnDisplay, lblLanguages

private void btnDisplay_Click(object sender, EventArgs e)
{
    string languages = "";

    if (chkMarathi.Checked) languages += "Marathi, ";
    if (chkEnglish.Checked) languages += "English, ";
    if (chkHindi.Checked) languages += "Hindi, ";
    if (chkSanskrit.Checked) languages += "Sanskrit, ";
    if (chkTamil.Checked) languages += "Tamil, ";

    if (languages.Length == 0)
        lblLanguages.Text = "No language selected.";
    else
        lblLanguages.Text = "Selected Languages: " +
                            languages.TrimEnd(',', ' ');
}
```

20. Create a Windows Forms application that changes the font size of a label “Hello, Welcome!” using a TrackBar. (Use TrackBar, Label Controls). CSMAP

```
// Controls: lblWelcome, trackBarFontSize
// Set lblWelcome.Text = "Hello, Welcome!"
// Set TrackBar Minimum = 8, Maximum = 48

private void trackBarFontSize_Scroll(object sender, EventArgs e)
{
    lblWelcome.Font = new Font(
        lblWelcome.Font.FontFamily,
        trackBarFontSize.Value,
        lblWelcome.Font.Style);
```

```
}
```